

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

CHE 315 — Past Questions & Materials
by Renance

Course: CHE 315
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 4
Levels: 200, 300

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Paper 1 of 4 — CHE 315 - 2024/2025 - First Semester - Petroleum Refining & Petrochemical Technology

Level 200 • First Semester • PastQuestion

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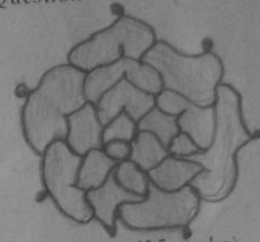
FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
 DEPARTMENT OF CHEMICAL ENGINEERING
 1ST SEMESTER EXAMINATION 2024/2025 044584 \$ 87
 ACADEMIC SESSION
 CHEM315/PTE 315: PETROLEUM REFINING & PETROCHEMICAL
 TECHNOLOGY

Instruction: 1) Answer all questions from section A and any one Question from section B.

2) Total duration- 3hours (Three hours)

SECTION A:

- 1a. What do you understand by the concept petroleum? (5marks)
- b. Describe the organic origin of petroleum formation (5mark)
- c. Crude oil pre-treatment before refining is necessary; explain with diagram. (15marks)
- 2a. Give the classes of products obtained from crude oil. (5marks)
- b. Using a technological diagram of single tower, explain the atmospheric distillation process of a crude oil. (15marks)
- c. Give five testing properties of petroleum product and their technical importance. (7marks)
- 3a. Give the classification of crude oil and explain why these are important in refining processes. (5marks)
- b. The layout of refining are divided into zones. What are the zones and its components? (10marks).
- c. Blending of fuels from different units of the same product are carried out. Explain the reasons for such operations. (10marks)



SECTION B

- 4a. What are the sources of petrochemical production, and the products obtain in their processes? (5marks)
 - b. Give the classification of plastics, and their characteristics. (5marks)
 - c. What the difference between synthetic fibres and elastomer. (5marks)
 - d. What are the common reactions that feature during petrochemical processes? (5marks)
 - 5a. Show polymerization processes, and the catalysts that are involved and the reaction sequence for ethylene conversion to polymer? (8marks)
 - b. By the help of a process diagram explain the production of low density polyethylene (LDPE). (12marks).
 - c. Show the stages of syngas production and give the significance of the product. (5marks)
 - 6a. What are the major constituent of natural gas? (5marks)
 - b. Explain the possible methods involved in natural gas purification. (5marks)
- Using a concise technological diagram describe "Selexol" process for acid gas oval. (15marks).

Paper 2 of 4 — CHE315/PTE315 PETROLEUM REFINING PETROCHEMICAL TECHNOLOGY 2025

Level 300 • PostUTME • PastQuestion

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THE DEPARTMENT OF CHEMICAL, ENGINEERING

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CHEMISPT M15; PETROL PLM REFINING & PETROCHEMICAL

TECHNOLOGY

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as per IN As.

; "Meher bo you'll understand the concept petroleum" (Smarter

2 Describe the organic origin of petroleum formation (5 marks)

Crude oil pre-treatment before refining is necessary: explain with diagram. (4 marks)

1. List the classes of products obtained from crude oil. (marks)

Using a schematic diagram of single tower, explain the atmospheric distillation process of a crude oil. (5 marks)

properties of petroleum product and their technical importance.

" Give the classification of crude oil and explain why these are important in refining

as the impurities of refining are divided into zones. What are the zones and its components?

(5 marks) ..

the blending of fuels from two different units of the same product are carried out. Explain the

process: FEESONS for the operation: (10 marks)

the 'cron B

the main sources of petroleum production, and the products obtained in their

the "processes? (5 marks) =

1. Give the classification of plastics and their characteristics. (5 marks)

What is the difference between synthetic fibres and elastomer. (5 marks)

2. What are the main chemical reactions that take place during petrochemical processes?

Paper 3 of 4 — CHE 315

Level 300 • First Semester • Handout

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①

Overview of Petrochemical Products

Most products from refinery are channeled into production of petrochemical products.

Product obtained from comes from the source.

1. Primary

Natural Gas

$\xrightarrow[\text{H}_2\text{O}]{\text{Catalytic}} \text{CO} + \text{H}_2$

$\xrightarrow{\text{for fertilizers}}$

Air

$\xrightarrow{\text{by air separation}}$

CO_2

N_2

He, Kr, ...

2 Intermediates

Thermal Cracking products such as

- ethane, propane, butane, Naphtha
- Cracked gases, such as olefins (ethylene, propylene, butylene, acetylene, etc.), liquid (benzene, toluene, xylene, etc.)

Olefins are the starting material for (monomers) for polyolefines plants. It also react with other hydrocarbons or non-hydrocarbon chemicals to generate vinyl chloride, ethylene glycol, neoprene, ethylene oxide etc, and subsequently used for as monomers for variety of polymers.

-ii-

Petroleum products converted to petrochemical Prod.

- Natural Gas (Gas oil field) — Carbon blacks, methanol, Acetic Acid, formaldehydes, Synthetic gas, chloromethanes, ethylene glycol, MTBE, acetylene, vinyl chloride.
- Ethane — Ethylene, polyethylene, ethyl alcohol, ethylene oxide, ethylene glycol, styrene, acrylonitrile, etc.
- Propane — Propylene, ~~butadiene~~, polypropylene, propylene oxide, isopropanol, acetone, etc.
- Butanes — Butadiene, acrylonitriles, polybutadiene, etc.
- Benzene — Phenol, styrene, polystyrene, cyclohexane, caprolactone, fibre, nylons, etc.
- Toluene — Nitrotoluene, phenol
- m-Xylene — p-Xylene
- p-Xylene — Tetracarboxylic acid
- O-Xylene — Phthalic acid.
- Sulfur — Sulfuric acid
- Wax — Microcrystalline wax, food grade cooking wax, cosmetic medical wax
- Coke — Metallurgical coke, electrode coke, etc.

- ii-
- Finished products of Petrochemicals:-
- Finished products from Petrochemicals have several varieties - rubbers, fibre, solvent, Paint, etc. The polymers obtained are used for Plastics, resins, lignol products, furniture components.
 - The plastics obtained are classified into
 - i - Thermoplastic (Thermoplast)
 - ii - Thermosetting (Thermoset)
 - The thermoplast are linear in molecular structure, and reshape by heating, or cooling. The heating and cooling doesn't change the properties of the structure.
 - Thermoset on the other hand on the other hand, by heating changes the structure of the network, either by crosslinking the molecule, and can not be reversed, after applying heat.
 - Synthetic fibres are family of plastic with very high modulus of elasticity as compare to plastics, and rubbers.
 - Rubbers & elastomer are polymers with low modulus of elasticity. It is used

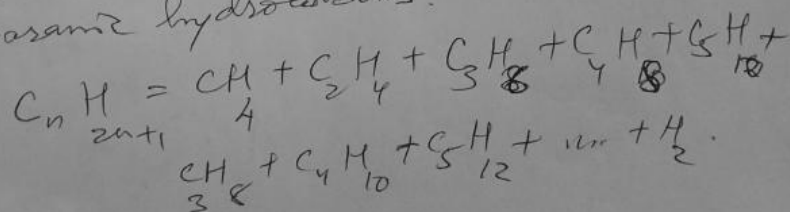
- iv -

with ~~used~~ sulfur to produce tire (process known as vulcanization).

Reactions featured during petrochemical processes includes:-

- i) catalyst, with:-
- ii) a) Dissociation — thermal/cat. crack process
 b) Dehydrogenation — "
 c) Hydrogenation
 d) Additional polymerization
 e) Condensation

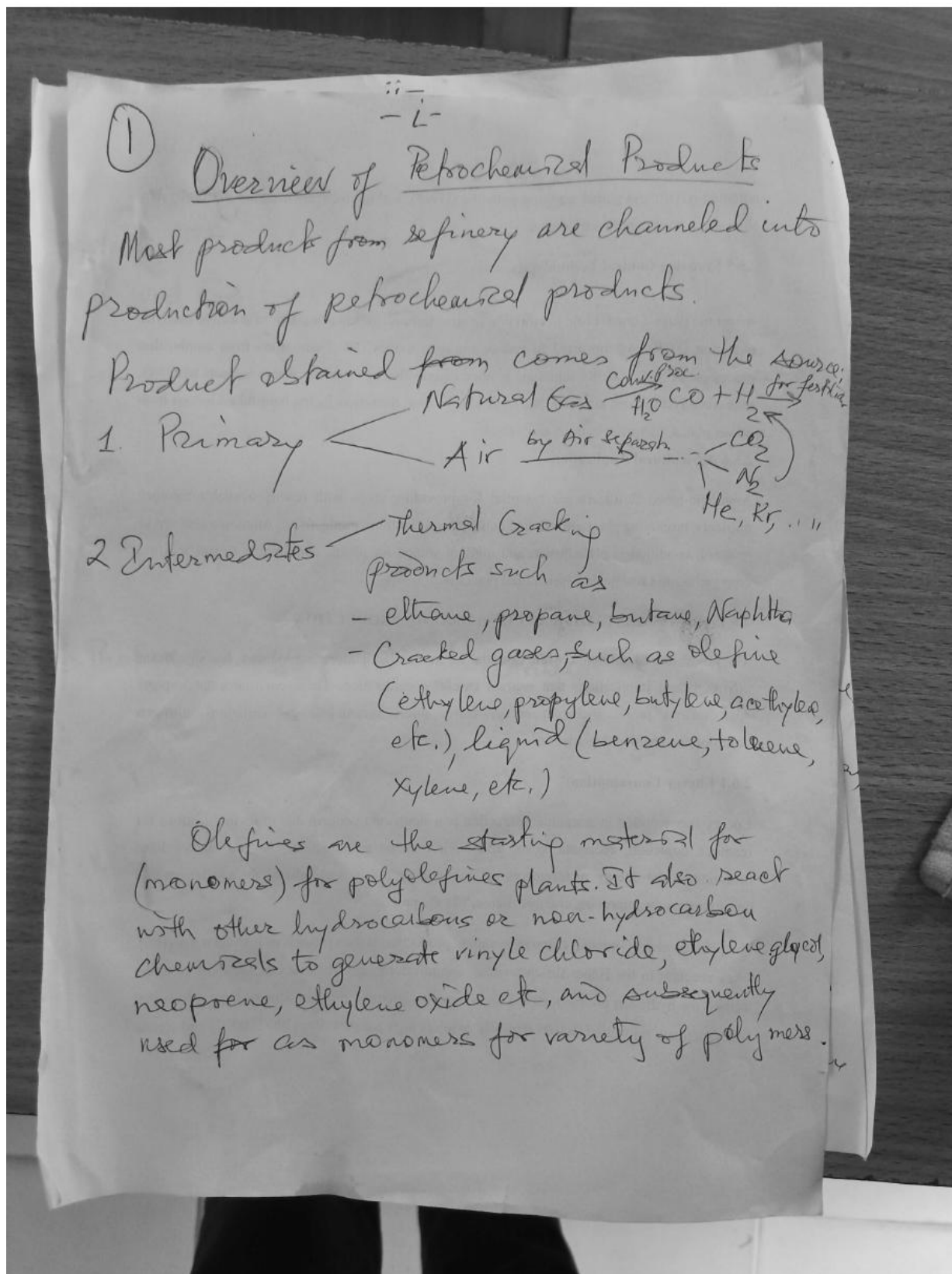
Naphtha - fraction initial IBP - 150, a low grade gasoline are sold to the petrochemical industry since it is rich in C5 - hydrocarbon, and can be cracked to obtain various homologous of paraffin hydrocarbons:-



Paper 4 of 4 — CHE 315

Level 300 • First Semester • Handout

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-ii-

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